(1)

(a) A matrix is said to be a hermitian if $A = A^{+}$

$$(AA^{+})^{+} = (A^{+})^{+} A^{+} = A A^{+}$$

as $(A^{+})^{+} = A$

$AA^{+}$ is hermitian

(b) $$(A - A^{+})^{+} = A^{+} - (A^{+})^{+}$$

$$= A^{+} - A = -(A - A^{+})$$

It is skew hermitian In this case we assumed $A \neq A^{+}$

2.

$$A = A^{+} \quad \& \quad B = B^{+}$$

$$(AB)^{+} = B^{+} A^{+} = BA$$

So, $AB$ is not hermitian

3.

$$\psi(\phi) = N \exp(-im\phi) \qquad 0 \leq \phi \leq 2\pi$$

$m =$ integer

The condition for Normalisation is

$$N^{*} N \int_{\phi=0}^{\phi=2\pi} \psi^{*}(\phi)\, \psi(\phi)\, d\phi = 1$$

$$\therefore \quad |N|^{2} \int_{0}^{2\pi} e^{im\phi} e^{-im\phi}\, d\phi = 1$$

$$|N|^{2}\, 2\pi = 1$$

$\therefore N = \sqrt{1/2\pi}$ (in general taken to be +ive)

4. (a) $\int_0^L \psi_1^* \psi_1 \, dx$

The integration $\int_0^L \psi^* \psi \, dx$ is finite,

$\psi$ can be normalized within 0 to L.

(b) $A^* A \int_0^L \sin\frac{\pi x}{L} \sin\frac{\pi x}{L} \, dx = 1$

$\therefore |A|^2 \int_0^L \sin^2\frac{\pi x}{L} \, dx = 1$

$\therefore \frac{|A|^2}{2} \int_0^L \left(1 - \cos\frac{2\pi x}{L}\right) dx = 1$

$\frac{|A|^2}{2}\left[\int_0^L dx - \int_0^L \cos\frac{2\pi x}{L} \, dx\right] = 1$

$\frac{|A|^2}{2} L - \left[\frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right)\right]_0^L = 1$

$= 0$

$\frac{|A|^2}{2} L = 1$

$\therefore \boxed{A = \sqrt{2/L}}$

So, the $\psi$ can be normalized within 0 to L and the normalization constant is $\sqrt{2/L}$

(In general we take +ive value as the negative value of A will not change the probability density $\psi^* \psi$)

(c) $$\hat{p}_x \psi(x) = -i\hbar \frac{\partial}{\partial x}\left(\sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)\right)$$

$$= -i\hbar \sqrt{2/L} \frac{\partial}{\partial x}\left(\sin \frac{\pi x}{L}\right)$$

$$= -i\hbar \sqrt{2/L} \cdot \left(\frac{\pi}{L}\right) \cos\left(\frac{\pi x}{L}\right)$$

$\therefore$ $\sqrt{2/L} \sin \frac{\pi x}{L}$ is not an eigenfunction

but $$\frac{\hat{p}_x^2}{2m} \psi(x) = \frac{(-i\hbar)^2}{2m} \left(\sqrt{2/L}\right)\left(\frac{\pi}{L}\right)^2 (-) \sin\left(\frac{\pi x}{L}\right)$$

$$= \frac{\hbar^2}{2m} \frac{\pi^2}{L^2} \sqrt{2/L} \sin\left(\frac{\pi x}{L}\right)$$

$$= \frac{\hbar^2 \pi^2}{2mL^2} \sqrt{2/L} \sin\left(\frac{\pi x}{L}\right)$$

$$= \frac{\hbar^2 \pi^2}{2mL^2} \sqrt{2/L} \sin \frac{\pi x}{L}$$

$$= \frac{\pi^2 \hbar^2}{2mL^2} \psi(x)$$

$\therefore$ $\psi(x)$ is an eigenfunction of K.E. operator and its eigenvalue $\frac{\pi^2 \hbar^2}{2mL^2}$

(e) $$\langle p_x \rangle = \frac{\int_0^L \psi^* \left(-i\hbar \frac{\partial}{\partial x}\right) \psi \, dx}{\int_0^L \psi^* \psi \, dx}$$

$$= 1$$

$$= \int \psi^* \left(-i\hbar \frac{\partial}{\partial x} \sqrt{2/L} \left(\sin \frac{\pi x}{L}\right)\right) dx$$

$$= \left(\sqrt{2/L}\right)^2 \int_0^L \sin\left(\frac{\pi x}{L}\right) \; -i\hbar \cos\left(\frac{\pi x}{L}\right) \cdot \frac{\pi}{L} \, dx$$

$$= \frac{2}{L} \cdot (-i\hbar) \frac{\pi}{L} \int_0^L \sin \frac{\pi x}{L} \cos \frac{\pi x}{L} \, dx$$

use $\boxed{2 \sin x \cos x = \sin 2x}$

$$\int_0^L \sin \pi x \cos \pi x \, dx = 0$$

$\therefore \boxed{\langle p_x \rangle = 0}$ average momentum is zero.

(f) Similarly we can prove

$\psi_2 = A \sin\left(\frac{2\pi x}{L}\right)$ is also an eigenfunction of K.E. operator with eigenvalue $\frac{4\pi^2\hbar^2}{2mL^2}$

(g) To prove

$$\int_0^L \psi_1^* \psi_2 \, dx = 0$$

$$\int_0^L \left(\sqrt{2/L}\right)^2 \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) dx$$

$$= \frac{2}{L}\int_0^L \sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{2\pi x}{L}\right) dx$$

$$= \frac{1}{L}\left[\int_0^L \cos\left(\frac{\pi x}{L} - \frac{2\pi x}{L}\right) - \cos\left(\frac{\pi x}{L} + \frac{2\pi x}{L}\right) dx\right]$$

$$= \frac{1}{L}\left[\int_0^L \cos\frac{\pi x}{L}\, dx - \int_0^L \cos\left(\frac{3\pi x}{L}\right) dx\right]$$

$$= 0$$

5. $e^{2ix}$ is an eigenfunction

(b) of $\frac{d^2}{dx^2}$ with eigenvalue

$(2i)^2 = -4$

(d) $Ae^{3ix} + Be^{-3ix}$ is

an eigenfunction with

$(3i)^2 = -9$

(e) $A\cos 4x + B\sin 4x$

is an eigenfunction with

eigenvalue 16

7

$$\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right](\cos ax)(\cos by)(\cos cz)$$

$$=\left(-a^2-b^2-c^2\right)\cos ax\cos by\cos cz$$

eigenfunction

8

$$A=\begin{pmatrix}0 & i\\ -i & 0\end{pmatrix}$$

a

$$A^{\dagger}=\begin{pmatrix}0 & i\\ -i & 0\end{pmatrix}$$

$A=A^{\dagger}$ hermitian

(b)

$$\begin{vmatrix} 0-\lambda & i \\ -i & 0-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - (-i^2) = 0$$

$$\lambda^2 - 1 = 0$$

$$\lambda = \pm 1$$

(a) $\phi$ = Non stationary and Normalized

$$\int \phi^* \phi \, dx = 1$$

$\psi_1$ is eigenfunction and $\{\psi_i\}$ are orthonormal

$$\therefore \int \psi_1^* \psi_1 \, dx = 1 = \int \psi_2^* \psi_2 \, dx$$

and $\int \psi_1^* \psi_2 \, dx = 0$

$$\int \phi^* \phi \, dx = 1$$

$$\therefore \int \left(\frac{1}{\sqrt{3}} \psi_1 + b \psi_2\right)^* \left(\frac{1}{\sqrt{3}} \psi_1 + b \psi_2\right) dx$$

$$\therefore \int \frac{1}{3} \psi_1^* \psi_1 \, dx + b^* b \int \psi_2^* \psi_2 \, dx + 0 + 0$$

$$\therefore \frac{1}{3} + |b|^2 = 1$$

$$|b|^2 = 2/3$$

$$b = \sqrt{2/3}$$

$\therefore$ average energy

$$= \sum_{i=1}^{2} |c_i|^2 E_i$$

$$= \left(\frac{1}{\sqrt{3}}\right)^2 E_1 + \left(\sqrt{2/3}\right)^2 E_2$$

$$= \frac{1}{3} \cdot 12 + \frac{2}{3} \cdot 24$$

$$= 4 + 16$$

$$= 20 \text{ eV}$$

Another way you can calculate

$$\int \phi^* \hat{H} \phi \, dx = \langle E \rangle$$

averge energy

$$= 20 \text{ eV}$$

(10)

$$\psi = N x \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

(a) It is normalizable and an acceptable function at $x = -\infty$ and $x = \infty$

In both extreme values of $x$

$\psi \to 0$, does not diverge

(b)

probability density

$$\psi^*\psi = |N|^2 x^2 \exp\left(-\frac{2x^2}{2r^2}\right)$$

$$\therefore \frac{d(\psi^*\psi)}{dx} = 0$$

$$2x \exp\left(-\frac{x^2}{r^2}\right) + \left[x^2 \exp\left(\frac{-x^2}{r^2}\right)(-)\frac{2x}{r^2}\right] = 0$$

$$\therefore 2x \exp\left(\frac{-x^2}{r^2}\right)\left[1 - \frac{x^2}{r^2}\right] = 0$$

$$x = 0, \quad x = \pm r$$

at $x = 0$ probability density $= 0$ (minimum)

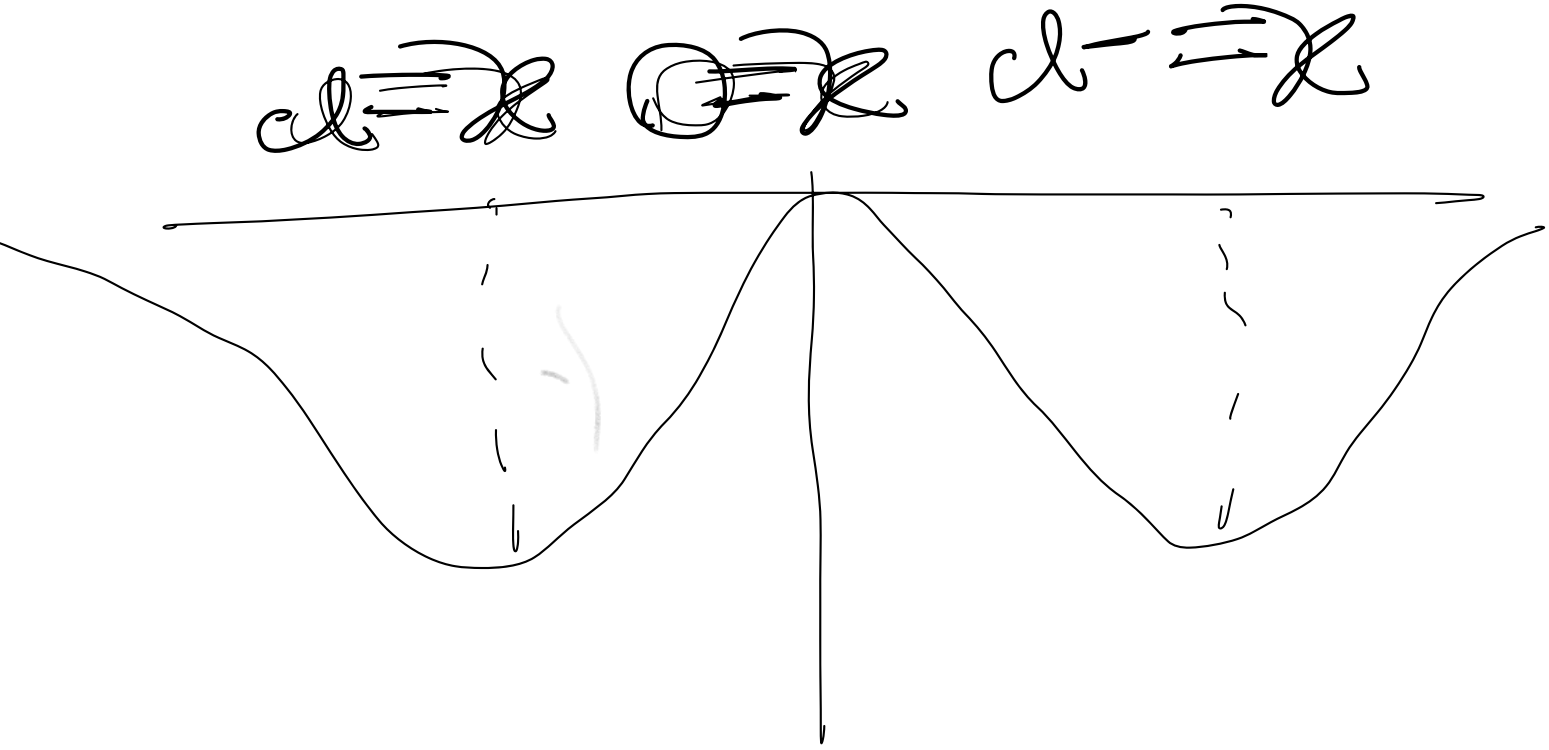
$x = \pm r$, the probability density will be maximum.

You can show,

$$\left.\frac{d^2(\psi^*\psi)}{dx^2}\right]_{x=r} < 0$$

and

$$\left.\frac{d^2(\psi^*\psi)}{dx^2}\right]_{x=-r} < 0$$

and

$$\left.\frac{d^2(\psi^*\psi)}{dx^2}\right]_{x=0} > 0$$